

Σ (3D)

from step 1

rotate

 $W \Sigma W^\top$

rotate into camera frame

 $J(\cdot)J^\top$ $\Sigma' = J W \Sigma W^\top J^\top$ 2×2 after cropping

clamp $t_x/t_z, t_y/t_z$ to $\pm \lim$
(step 5)

$$J = \begin{pmatrix} f_x/t_z & 0 & -f_x t_x/t_z^2 \\ 0 & f_y/t_z & -f_y t_y/t_z^2 \\ 0 & 0 & 0 \end{pmatrix}$$

$$f_x = \frac{W_{\text{img}}}{2 \tan(\text{fov}_x/2)}, \quad f_y = \frac{H_{\text{img}}}{2 \tan(\text{fov}_y/2)}$$

W : rotation block of the view matrix; J : affine (local) approximation at μ